

Linear Algebra — Exercises

Vector spaces

Exercise 1. Determine whether or not the following sets are vector spaces over the indicated fields:

- (a) $\mathbb{Z}$ over $\mathbb{Z}_2$, with addition of integers, and scaling defined as $0 \cdot z = 0$, $1 \cdot z = z$ for every $z \in \mathbb{Z}$
- (b) $\mathbb{Z}_4$ over $\mathbb{Z}_2$, with addition modulo 4, and scaling defined as $0 \cdot z = 0$, $1 \cdot z = z$ for every $z \in \mathbb{Z}_4$
- (c) $\mathbb{C}$ over $\mathbb{R}$, with addition of complex numbers, and scaling by real numbers
- (d) $\mathbb{R}$ over $\mathbb{C}$, with addition of real numbers, and scaling by complex numbers
- (e) all subsets of a finite set X over $\mathbb{Z}_2$, with the symmetric difference as vector addition, and with scaling defined as $0 \cdot A = \emptyset$ and $1 \cdot A = A$ for every $A \subseteq X$
- (f) $\mathbb{R}_+$ over $\mathbb{R}$, with addition of real numbers, and scaling defined as $p \odot r = r^p$ for every $r \in \mathbb{R}_+$ and $p \in \mathbb{R}$

Exercise 2. Decide which of the following subsets are subspaces:

- (a) $\{(x, y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0\} \subseteq \mathbb{R}^2$
- (b) $\{(x, y, z) \in \mathbb{R}^3 \mid yz \leq 0\} \subseteq \mathbb{R}^3$
- (c) $\{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = x - y = 0\} \subseteq \mathbb{R}^3$
- (d) $\{(x, y) \in \mathbb{R}^2 \mid |x - y| \leq 1\} \subseteq \mathbb{R}^2$

Exercise 3. Prove that in a vector space V over a field $\mathbb{K}$, for every $\mathbf{v}, \mathbf{u} \in V$ and $\lambda \in \mathbb{K}$,

$$\lambda \cdot (\mathbf{v} - \mathbf{u}) = \lambda \cdot \mathbf{v} - \lambda \cdot \mathbf{u}.$$

Exercise 4. The intersection of any collection of subspaces of a vector space V is a subspace of V . Show that this is not necessarily true for the union of subspaces.

Exercise 5. The real plane $\mathbb{R}^2$ is a vector space over $\mathbb{R}$. Describe, in geometrical terms, all subspaces of $\mathbb{R}^2$.

Exercise 6. Let V be a vector space over a field $\mathbb{K}$ and let $S, T \subseteq V$. Prove that:

- (a) $S \subseteq \text{span}(S)$
- (b) $S \subseteq T \implies \text{span}(S) \subseteq \text{span}(T)$
- (c) $\text{span}(\text{span}(S)) = \text{span}(S)$
- (d) $\mathbf{v} \in \text{span}(S) \iff \text{span}(S) = \text{span}(S \cup \{\mathbf{v}\})$.